CU GEODATA

EST. 2020

SPONSORSHIP

2025-2026

Cornell University

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ABOUT US

Team Background

CU GeoData is a student-run project team at Cornell. We design, build, and deploy instrumentation capable of recording a large variety of atmospheric, geologic, and hydrologic data. We are currently collaborating with a number of different organizations, such as Cornell faculty and the Paleontological Research Institute. These partners provide resources and mentorship, allowing us to generate accurate data and environmental assessments.

Our Work and Impact

Our team utilizes an interdisciplinary connection between engineering and the Earth and atmospheric sciences. We combine our skills, knowledge, and networks to form a deep understanding of natural systems, putting us in a unique position to addressing complex environmental issues. Through our innovative and collaborative work, CU GeoData contributes to the development of sustainable solutions that will help to protect our planet.

TEAM LEADERSHIP

KYLIE MILLER
PRIMARY LEAD

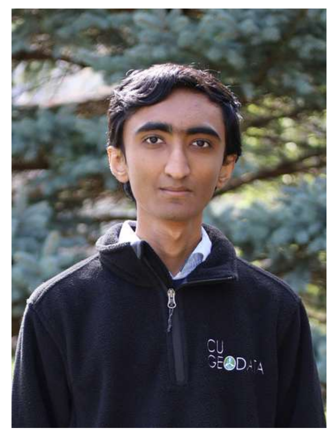
ORION HOCH
SECONDARY LEAD

NOUR KASTOUN
SENIOR ADVISOR

HENRY MU
WATER LEAD

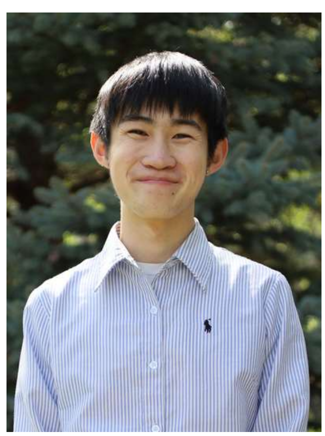
AYA SAUTE
ROCK LEAD

JACK HALBERSTADT
AIR LEAD



AVINASH ARAVIND
TECH LEAD

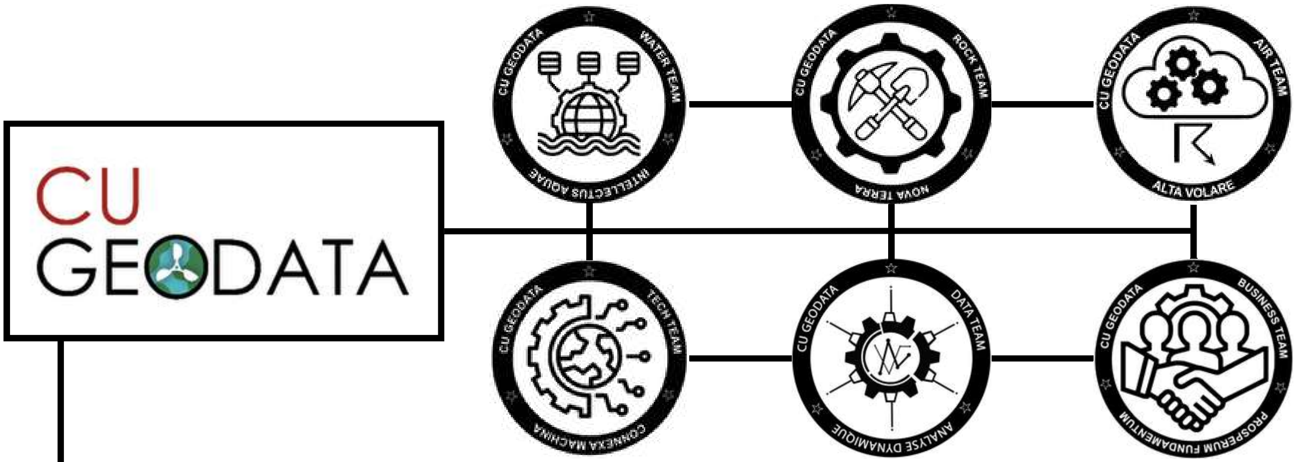
RITHYA SRIRAM
DATA CO-LEAD



WILLIAM LEUNG
DATA CO-LEAD

MEET BAANI KAUR
BUSINESS LEAD

FACULTY COLLABORATORS



DR. DAVID HYSELL

As CU GeoData's faculty advisor, Dr. Hysell meets with Team Leads on a weekly basis. He also provides guidance on complex technical issues during instrumentation development cycles and supports the Tech Subteam with circuitry expertise.

DR. LOU DERRY

Dr. Derry offers his rich experiences as a biogeochemist to CU GeoData's Rock Subteam. Dr. Derry has been invaluable in engaging CU GeoData members with laboratory skills and mine tailings analysis. Dr. Derry has also worked with CU GeoData's Water Subteam on past projects.

DR. TOBY AULT

As an expert in weather seasonality, drought, and timescales of climate change, Dr. Ault has provided valuable insight and guidance for our weather and soil projects. CU GeoData is engaged in an ongoing collaboration with the Emergent Climate Risk Lab to install a low-cost air quality and soil moisture sensor in partnership with community stakeholders.

DR. ARTHUR DEGAETANO

Dr. Degaetano’s vast expertise and interest in applied climatology and meteorology, as well as his dedication to community outreach, make him an important mentor to CU GeoData's Air Sub-team. Through this partnership, CU GeoData members have access to the Game Farm field site and are able to share weather station data on the Network for Environment and Weather Application (NEWA) website.

DR. ROWENA B LOHMAN

Dr. Lohman, who is the soil moisture lead on the NASA-ISRO Synthetic Aperture Radar, has shared her experience and expertise in cutting-edge remote sensing technology with the CU GeoData Rock Team. Dr. Lohman has been a valuable resource to the Rock Team during their soil moisture sensor projects.

AIR SUB-TEAM

Tethersonde

Our longest lasting and ongoing project is the tethersonde project. We send a student-built tethersonde up into the lowest 500 feet of the atmosphere to measure how atmospheric variables such as temperature, humidity, wind speed, and wind direction change with height. Designed for affordability, mobility, and education, the system enables high-resolution observations across many environments. We are able to characterize boundary-layer structure during Finger Lakes lake-effect snow events, capture spatial variations in wind and temperature across local terrain, and support studies of urban-rural contrasts, sea breezes, and convective development.

Weather Stations

We are currently expanding local weather station networks in order to collect more surface meteorological data to better forecast Finger Lakes precipitation.

3D Printing

To connect with the broader scientific community, Air Team has begun building a 3D-printed weather station in an effort to lead the way in making weather stations and data collection more affordable and accessible in the future.

Air Sub-team Members

Weather Station Installation

Tethersonde Deployment

Sensor Maintenance

CU GEODATA
AIR TEAM
ALTA VOLARE

ROCK SUB-TEAM

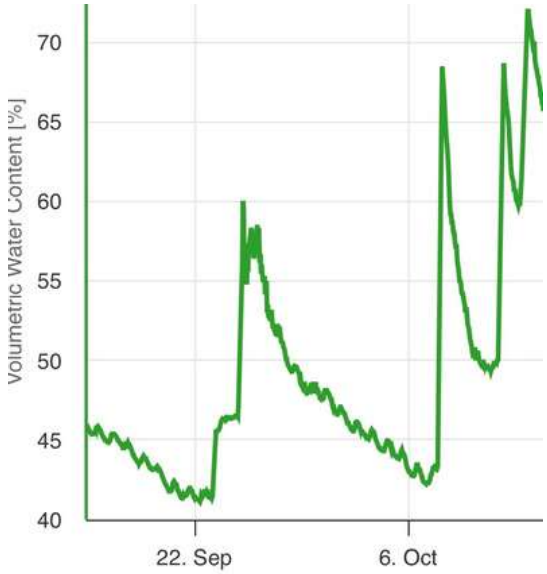
Empire State Mine Tailings Impact Study

With Professor Derry's lab, we determined the elemental composition of mine tailing samples using inductively coupled plasma mass spectrometry over a time series from 2009 to 2019. We also used satellite imagery to develop maps to analyze other environmental factors, including precipitation and air quality. Our goal is to track the local environmental impacts of changes in mine ownership.

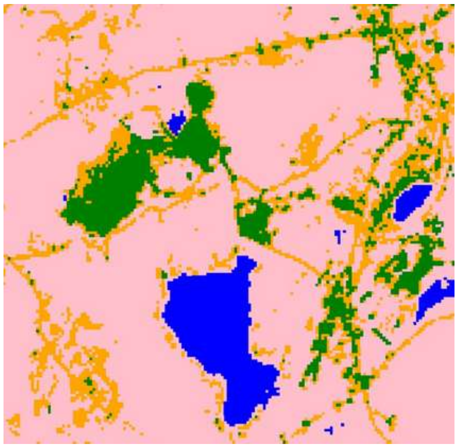
Team-Wide Sensor Network

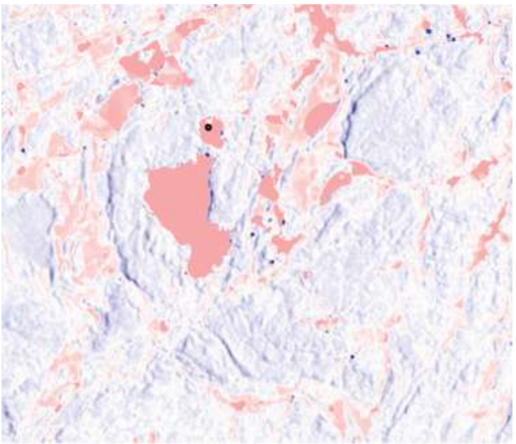
Rock Team, in partnership with local sensor developers and community organizations, is installing a low-cost network of 17 soil moisture sensors and five air quality sensors around Cayuga Lake. Five soil moisture sensors at the Game Farm field site are currently collecting data to ground-truth the NISAR satellite's soil moisture algorithms. The remaining sensors will be installed in the next two months in preparation for a team-wide lake effect and productivity study.

Rock Sub-team Members



Soil Volumetric Water Content Over Time





Land Classification Map and Soil Moisture Map

WATER SUB-TEAM

Cayuga Buoy Network

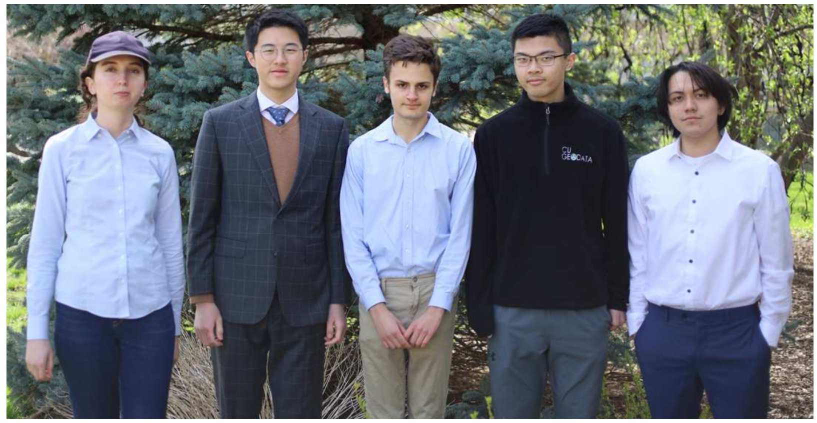
We are currently working on creating a fleet of buoy weather stations in order to study local Lake Effect Snow across Cayuga Lake. We are using data to increase the accuracy of our predictions.

Sap Flow Sensors

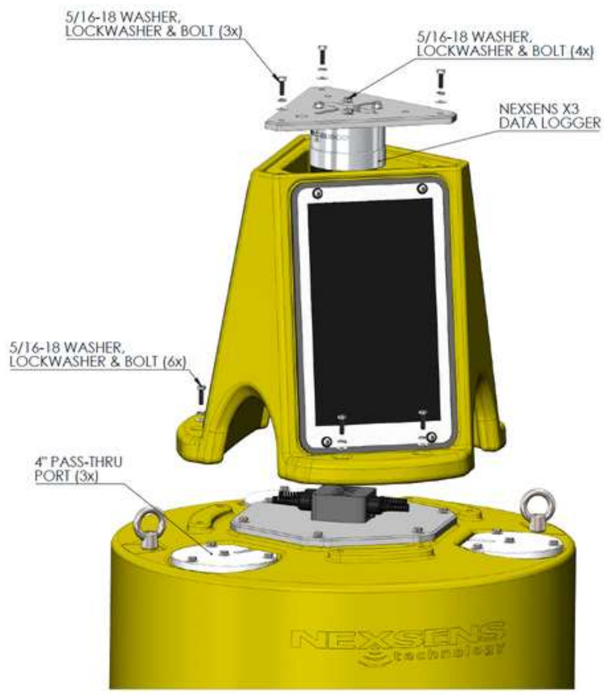
Water team is planning the implementation of sap flow sensors to measure maple transpiration and overall tree health. These sensors will be placed around CNC to study the climatological tendencies of the area through the tree proxy.

Data Loggers

We are also experimenting with Internet of Things tech in order to stream the data from our sensors to online servers more efficiently and cost effectively.



Water Sub-team Members



Weather Station Buoy
Source: Nexsens Technology

CU GEODATA WATER TEAM INTELLECTUS AQUAE

TECH SUB-TEAM

Water Sensors

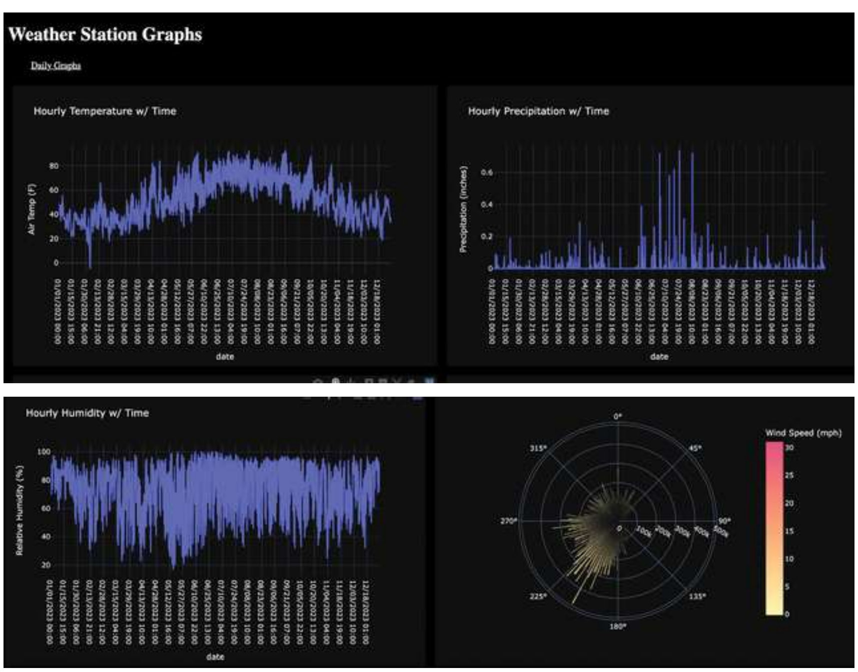
The goal of our current water sensor project is to monitor lake conditions leading to harmful algae blooms. We are interested in measuring parameters such as temperature, pH, total dissolved solids (TDS), turbidity, and pressure. We aim to report data live to a local server through an SMS connection.

Drone

We are using a drone equipped with an infrared camera to create a normalized differential vegetation index (NDVI), which is used to measure vegetation health. We have continued the drone project by making updates to firmware, leveling flight arms, and recalibrating sensors. Our current goal is to equip the drone with environmental sensors for aerial data collection.

Tech Team Members

Drone



Weather Station Plots Using HTML and Python

SPONSORSHIP LEVELS

Each upper level includes all features of the levels below it

LEGACY: $3,500+

- An Onset weather station or Nexsens weather station buoy will be purchased and deployed in the Ithaca community.

GOLD: $2,500

- Representatives from your organization or company have the option of joining priority meetings with team members to discuss our projects.
- CU GeoData can host information sessions about your company that are open to members and the greater Cornell community.

SILVER: $1,000

- You will receive a semesterly newsletter with updates on our team and projects.

BRONZE: $500

- You will have access to the GeoData team member resume book for recruiting and networking.

WHY CONTRIBUTE?

During our beginning years, CUGeoData was funded by the Cornell Shen Fund for Social Impact, but we have big plans! We need your support in order to continue making impactful scientific discoveries for the long-haul. You can help us achieve our goals and complete every project proposal.

CONTRIBUTIONS ENABLE US TO:

- **Make meaningful scientific discoveries**
- **Help our community and local environment**
- **Engage students in hands-on fieldwork**
- **Gain leadership and technical skills**
- **Develop problem solving skills with real-world science and engineering applications**

SPONSORSHIP PERKS:

- **Increase your recruiting presence on campus**
  Through direct access to CU GeoData members and recruiting events
- **Access to the CU GeoData Resume Book**
  View members' credentials and relevant experience for recruitment
- **Branding and PR**
  CU GeoData will advertise your company on our merchandise, equipment, presentations, plaques, etc
- **Tax Deductible Contributions**
  Your donations to CU GeoData will be tax deductible

Contact Us

For more details on our work and to learn more about the impact of your contribution

Email our Business sub-team :

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View our website: cugeodata.com

Connect with us on social media:

@cugeodata

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Cornell Engineering | Cornell CALS College of Agriculture and Life Sciences

Earth and Atmospheric Sciences